Master of Science in Bionics Engineering

Academic Year 2025-2026

Pontedera, March 19, 2026

Bioinspired and Soft Robotics

Soft Robotics Technologies

Actuation Technologies

Introduction

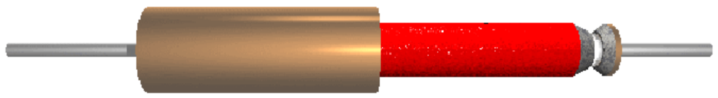
Matteo Cianchetti - matteo.cianchetti@santannapisa.it

Some definitions…

Actuator: system which is able to provide a mechanical action in the requested time interval.

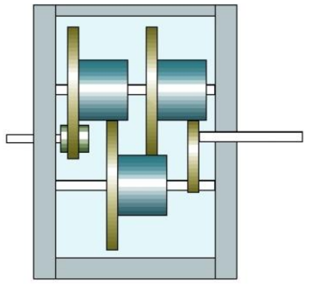
An actuator is composed of:

- A **transducer**: physical element which converts any kind of energy input (typically electrical) in mechanical energy - MANDATORY

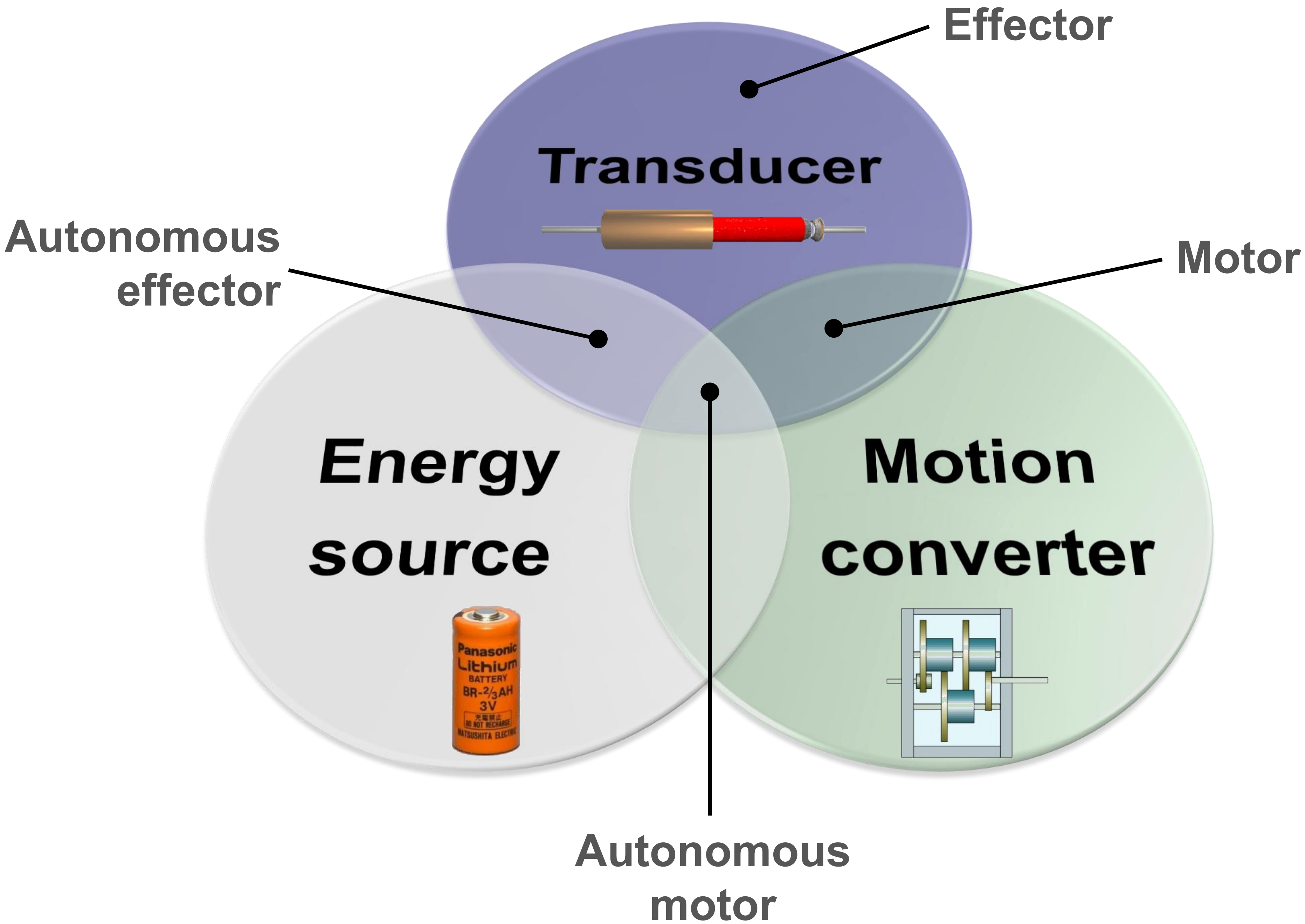


- An **energy source**: element which provides electric energy for the transformation into mechanical energy - OPTIONAL

- A **motion converter**: mechanism which transforms a mechanical coupling (force_1 with speed_1) into another (force_2 with speed_2) - OPTIONAL



Some definitions…



The analysis of an actuator cannot be limited to the transducer performances. An actuator is a SYSTEM where energy source and motion converter have key roles.

Fundamental characteristics

- STRESS: typical force exerted on area unit (maximum value is evaluated as maximum static force achievable divided by area unit)

$$\sigma = \frac{F}{A} \quad \left[\frac{N}{m^2} = Pa\right]$$

- STRAIN: deformation normalized respect to initial length on actuation direction (maximum value is evaluated using the maximum deformation which is NOT necessarily the maximum stress condition)

$$\varepsilon = \frac{l_f - l_0}{l_0} \quad \left[\frac{m}{m}\right]$$

- SPECIFIC WORK: generated work per mass unit of the actuator (maximum value is evaluated by dividing the maximum product between stress and strain with the density)

$$W_s = \frac{\sigma\varepsilon}{\rho} \left[\frac{N}{m^2}\frac{m}{m}\frac{m^3}{kg} = \frac{J}{kg}\right]$$

- SPECIFIC POWER: generated power per mass unit of the actuator (maximum value is evaluated by dividing the maximum power with the actuator mass)

$$P_s = \frac{P}{m} \quad \left[\frac{W}{kg}\right]$$

Fundamental characteristics

- FREQUENCY: number of cycles the actuators is able to perform in the time unit

$$f = \frac{n^\circ cycles}{s} \quad [Hz]$$

- POWER DENSITY: ratio between the effective power and the actuator volume

$$P_d = \frac{1}{2}\sigma\varepsilon f \quad \left[\frac{N}{m^2}\frac{m}{m}\frac{1}{s} = \frac{W}{m^3}\right]$$

- TRANSDUCER EFFICIENCY: ratio between the mechanical work produced and the energy input

$$\eta_t = \frac{W_o}{E_i} \quad \left[\frac{J}{J}\right]$$

- MOTION CONVERTER EFFICIENCY: ratio between the output and the input mechanical energy

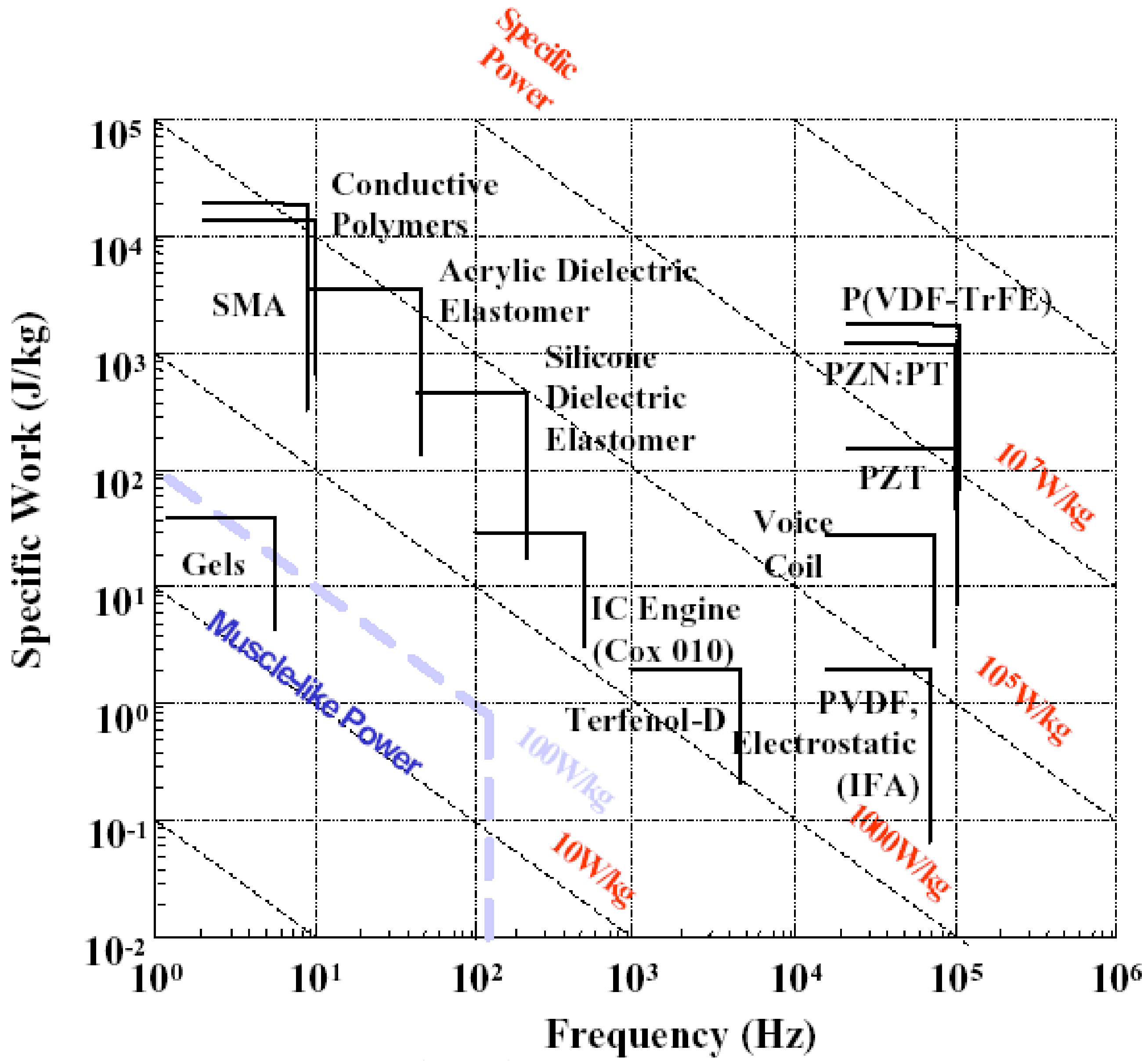
$$\eta_{mc} = \frac{E_o}{E_i} \quad \left[\frac{J}{J}\right]$$

- BULK RATIO: volume ratio between the motion converter and the transducer

$$BR = \frac{V_{mc}}{V_t} \quad \left[\frac{m^3}{m^3}\right]$$

However…

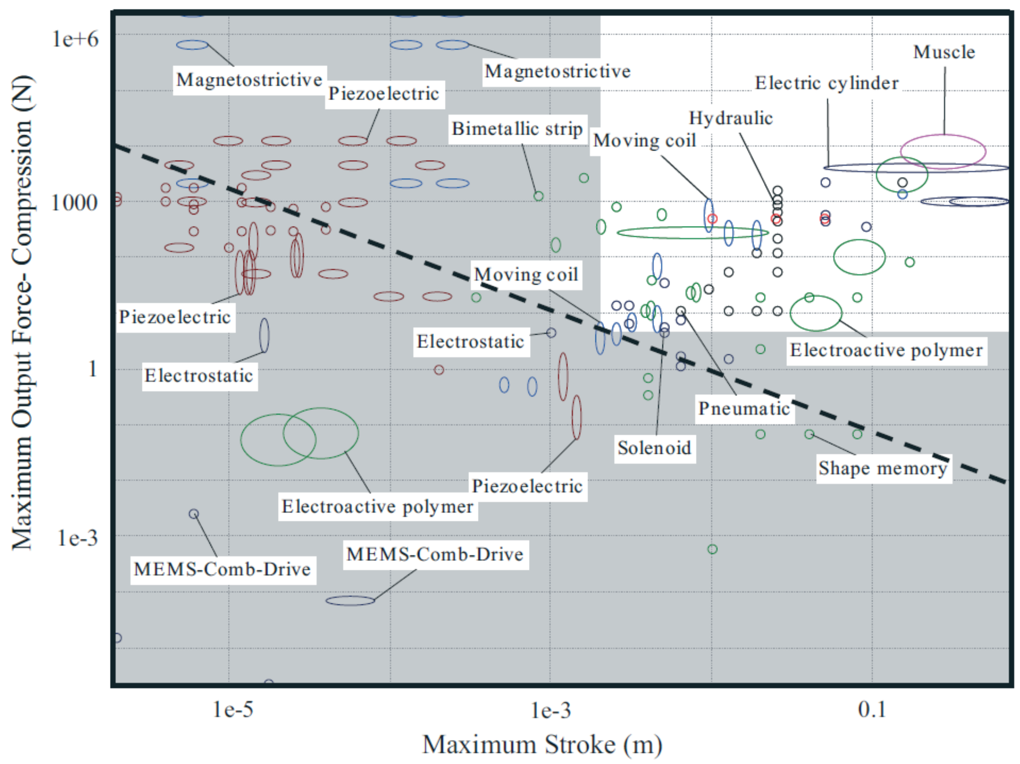
… they are rarely all compared

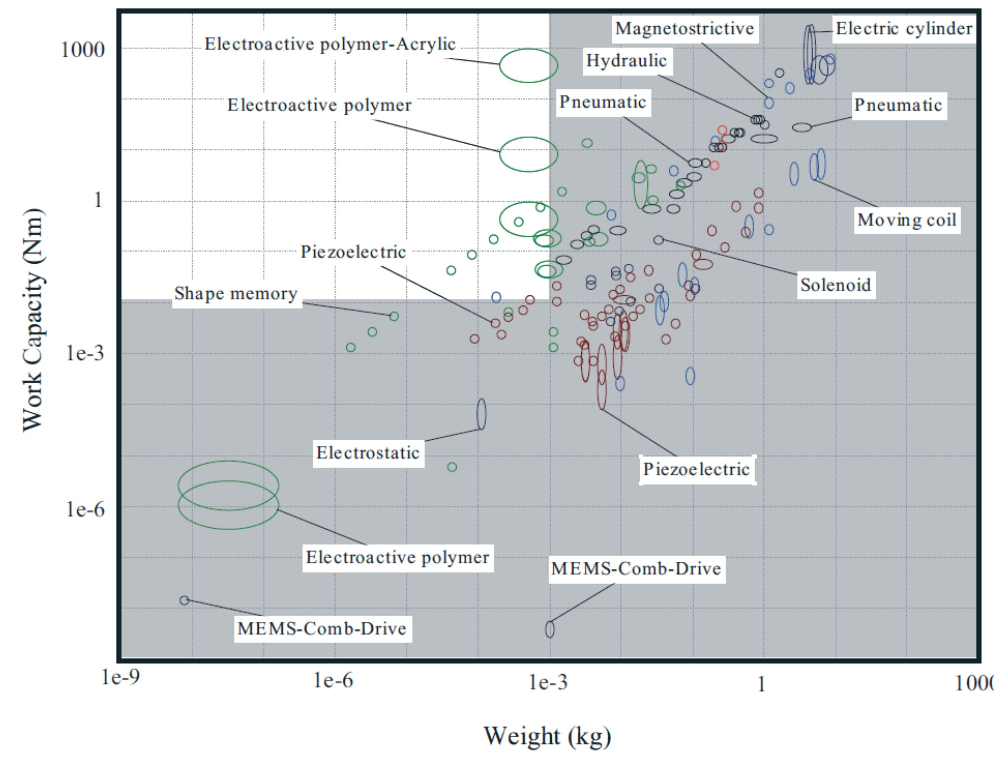


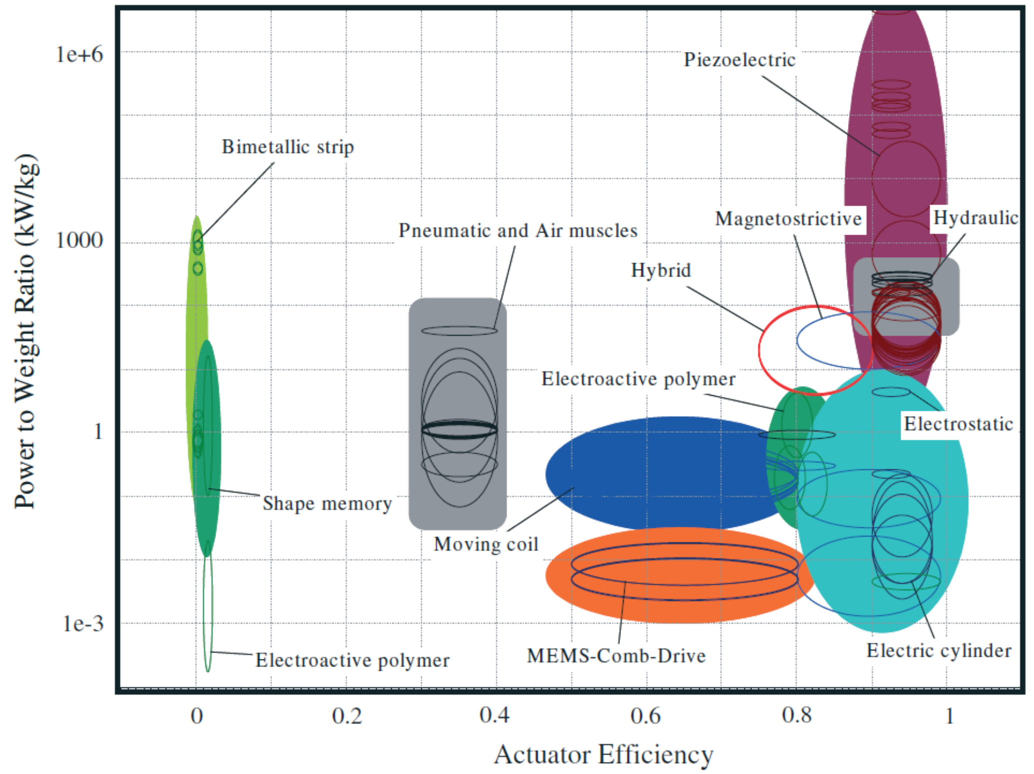
Source: DARPA and SRI International

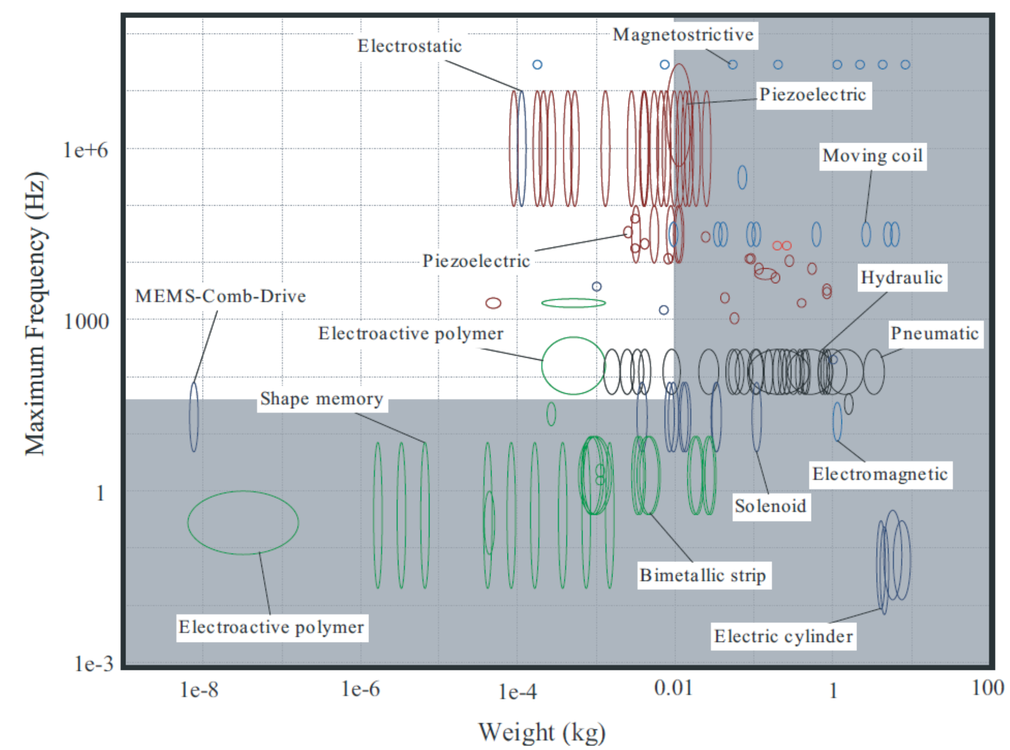
This comparison makes sense only if motion converter and energy source are neglected

Ashby diagrams









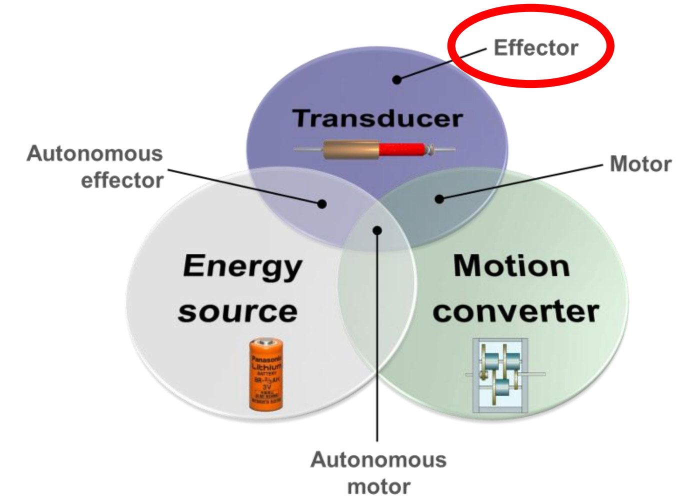
Comparison tables

Well presented, easy to read, but use carefully

	Stress (MPa)	Strain	Efficiency	Bandwidth (Hz)	Power (W/cm3)
Electromagnetic	0.02	0.5	90%	20	0.1
Hydraulic	20	0.5	80%	4	20
Pneumatic	0.7	0.5	90%	20	3.5
Muscle	0.35	0.2	30%	10	0.35
Shape Memory	200	0.1	3%	3	30
Electrostrictive	50	0.002	50%	5000	250
*Piezoelectric	35	0.002	50%	5000	175
Magnetostrictive	35	0.002	80%	2000	70
Contractile polymer	0.3	0.5	30%	10	0.75
*Single Crystal PZN:PT	300	0.017	90%	5800	15000

Pons, J.L., "Emerging Actuator Technologies: A Micromechatronic Approach", John Wiley & Sons, Ltd, England, 2005

Know what you need



Effector

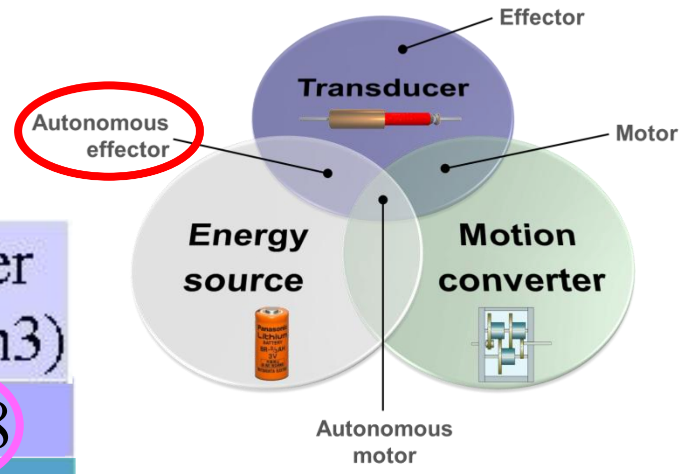
	Stress (MPa)	Strain	Efficiency	Bandwidth (Hz)	Power (W/cm3)
Electromagnetic	0.02	0.5	90%	20	0.1
Hydraulic	20	0.5	80%	4	20
Pneumatic	0.7	0.5	90%	20	3.5
Muscle	0.35	0.2	30%	10	0.35
Shape memory	200	0.1	3%	3	30
Electrostrictive	50	0.002	50%	5000	250
Piezoelectric	35	0.002	50%	5000	175
Magnetostrictive	35	0.002	80%	2000	70
Contractile polymer	0.3	0.5	30%	10	0.75
*Single Crystal PZN:PT	300	0.017	90%	5800	15000

Not considered

- Unlimited energy source
- Transducer efficiency not considered

$$P_{d,effector} = \frac{1}{2}\sigma\varepsilon f$$

Know what you need



Autonomous effector

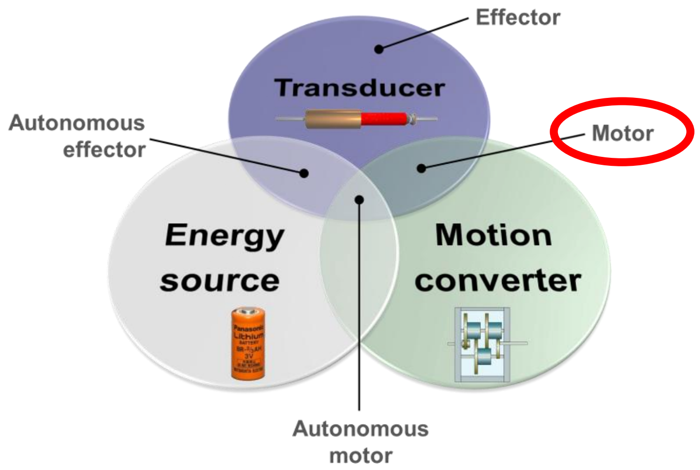
	Stress (MPa)	Strain	Efficiency	Bandwidth (Hz)	Power (W/cm3)
Electromagnetic	0.02	0.5	90%	20	0.08
Hydraulic	20	0.5	80%	4	16
Pneumatic	0.7	0.5	90%	20	3.15
Muscle	0.35	0.2	30%	10	0.1
Shape memory	200	0.1	3%	3	0.01
Electrostrictive	50	0.002	50%	5000	125
Piezoelectric	35	0.002	50%	5000	0.2
Magnetostrictive	35	0.002	80%	2000	56
Contractile polymer	0.3	0.5	30%	10	0.23
*Single Crystal PZN:PT	300	0.017	90%	5800	13500

- Limited energy source
- Transducer efficiency considered
- Battery power density considered

$$P_{d,aut.effector} = \frac{P_{d,effector}}{1 + \frac{1}{\eta_t}\frac{P_{d,effector}}{P_{d,battery}}}$$

Know what you need

Motor

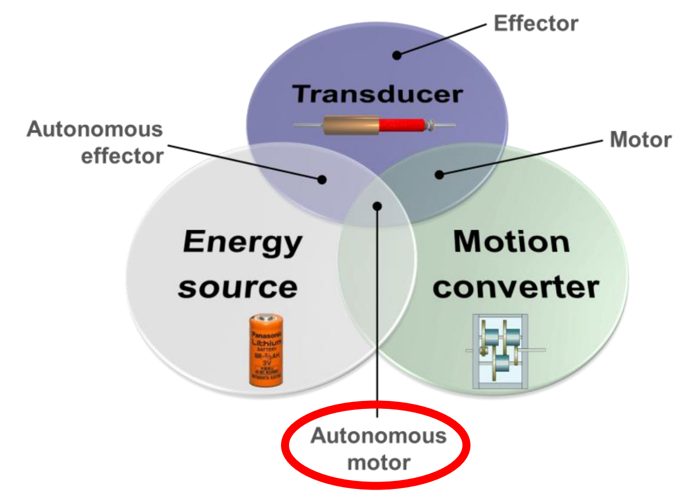


	Motion Converter Bulk Ratio	Motion Converter efficiency (%)	Motor Power (W/cm³)
Electromagnetic	1	80	0.04
Shape Memory	10	50	1.4
Piezoelectric	100	30	0.52

- Unlimited energy source
- Transducer efficiency not considered
- Bulk ratio considered

$$P_{d,motor} = \frac{P_{d,effector}}{1+BR}\eta_{mc}$$

Know what you need



Autonomous motor

	Motion Converter Bulk Ratio	Motion Converter efficiency (%)	Auton. Motor Power (W/cm^3)
Electromagnetic 90%	1	80	0.035
Shape Memory 3%	10	50	0.006
Piezoelectric 50%	100	30	0.054

- Limited energy source
- Transducer efficiency considered
- Battery power density considered
- Bulk ratio considered

$$P_{d,aut.motor} = \frac{P_{d,effector}}{1 + \frac{1}{\eta_t}\frac{P_{d,effector}}{P_{d,battery}} + BR}\eta_{mc}$$

What do you need?

Requirements:

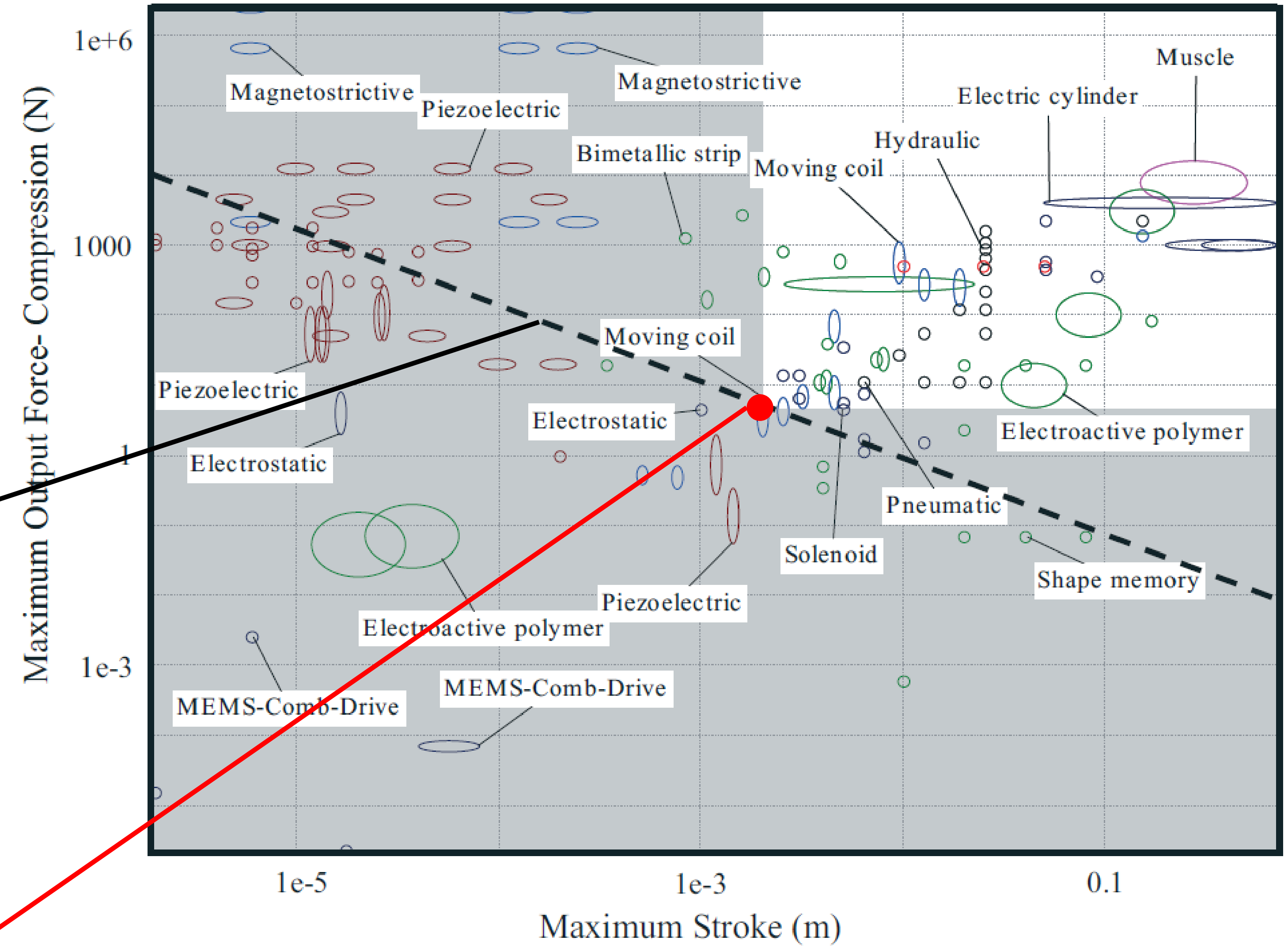
Force	5 N
Stroke	2 mm
Frequency	50 Hz
Life	At least 10^8 cycles
Mass	As light as possible

What do you need?

Requirements:

Force	5 N
Stroke	2 mm
Frequency	50 Hz
Life	At least 10^8 cycles
Mass	As light as possible

Nothing below this line can be acceptable



Force x Stroke = Work capacity

5N x 0.002m = 0.01Nm

Possible solutions:

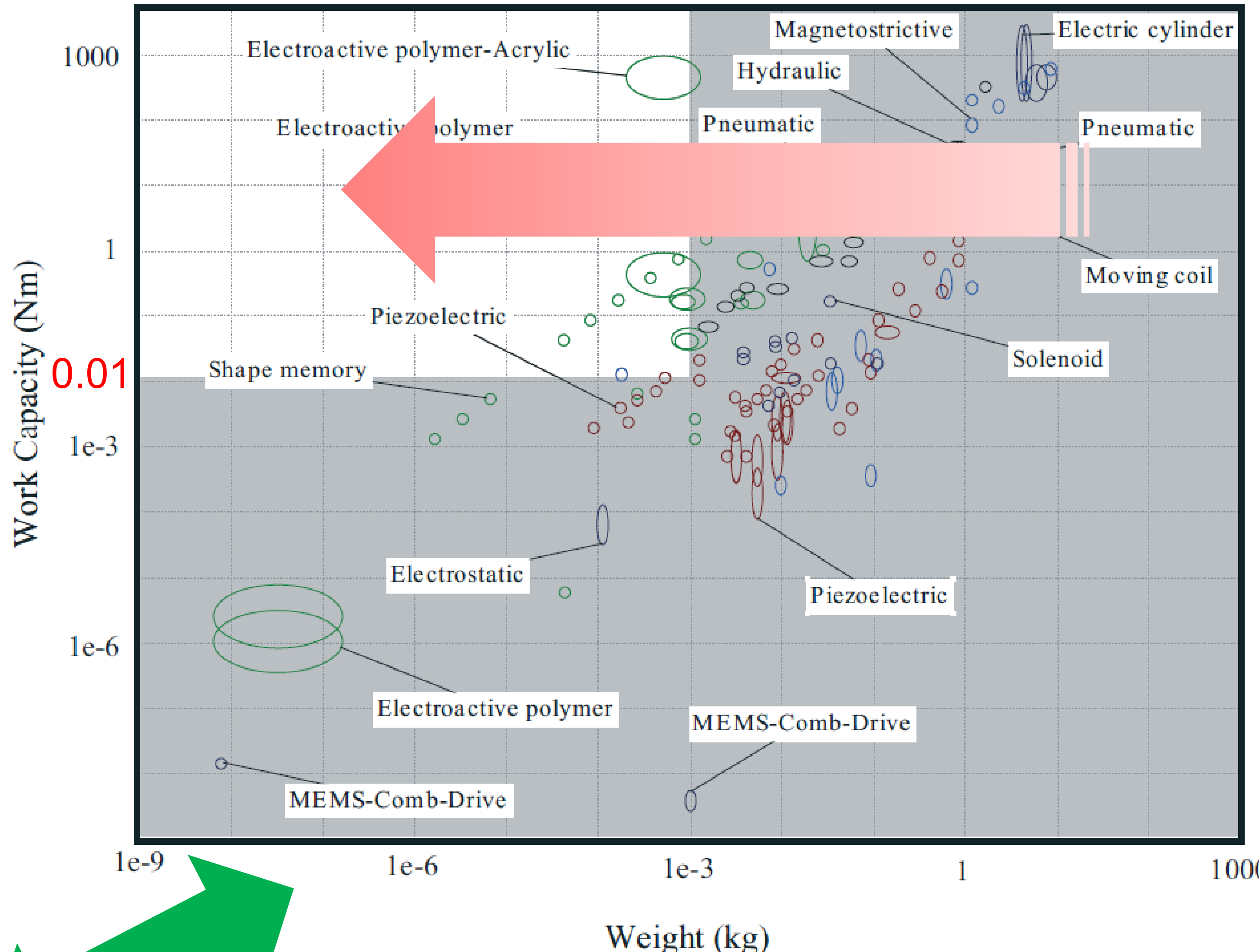
Work capacity ≥ 0.01Nm

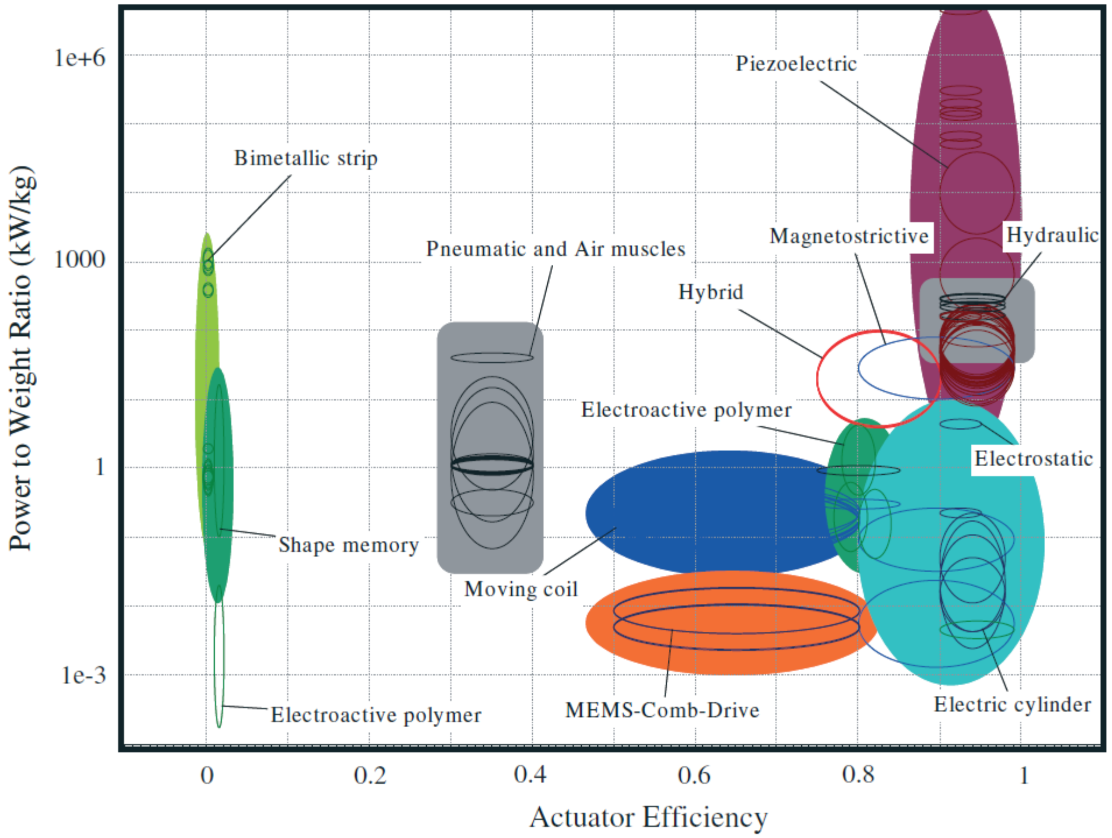
(Only if levers, gears or screw mechanisms are allowed)

What do you need?

Requirements:

Force	5 N
Stroke	2 mm
Frequency	50 Hz
Life	At least 10^8 cycles
Mass	As light as possible



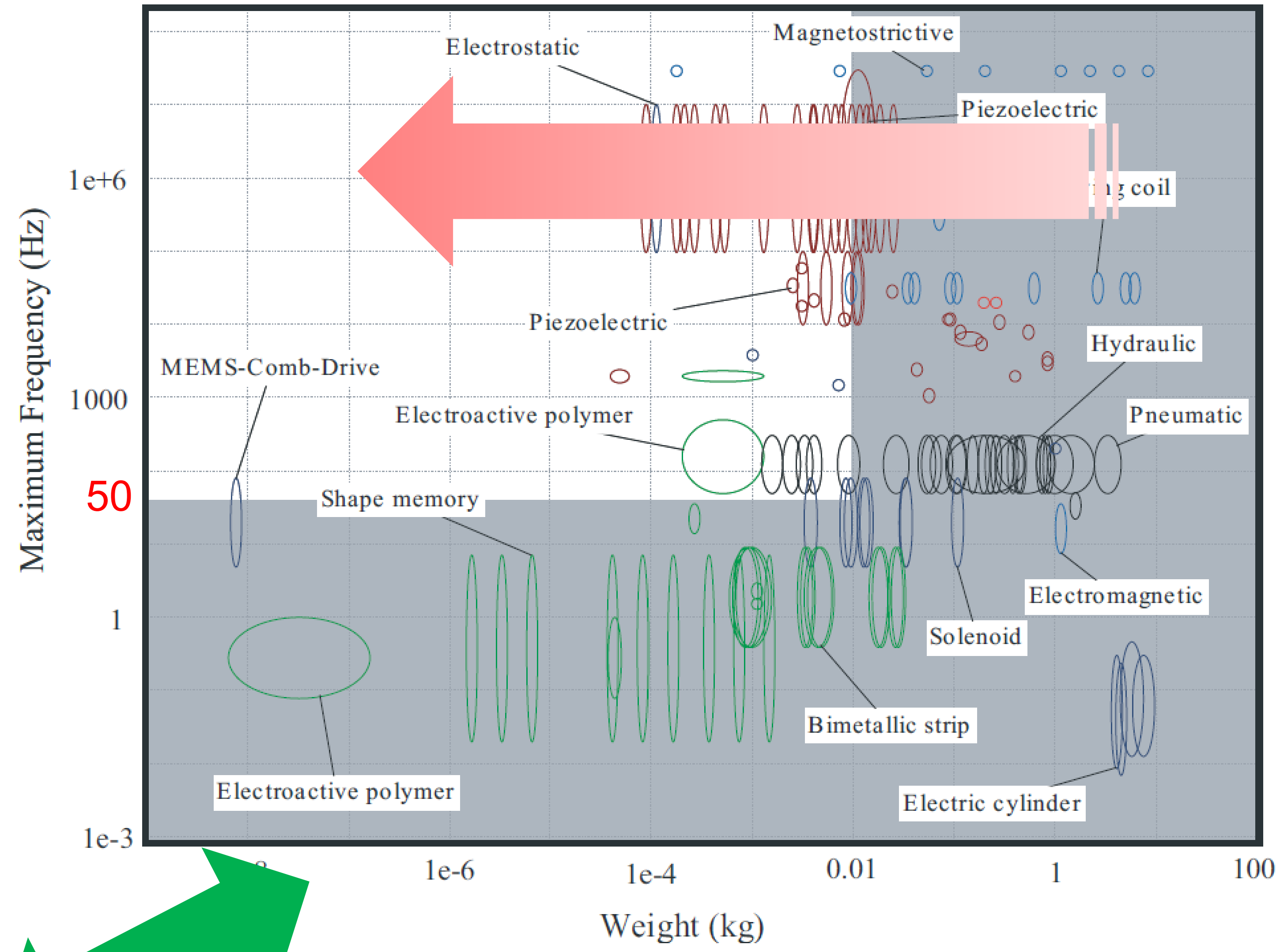


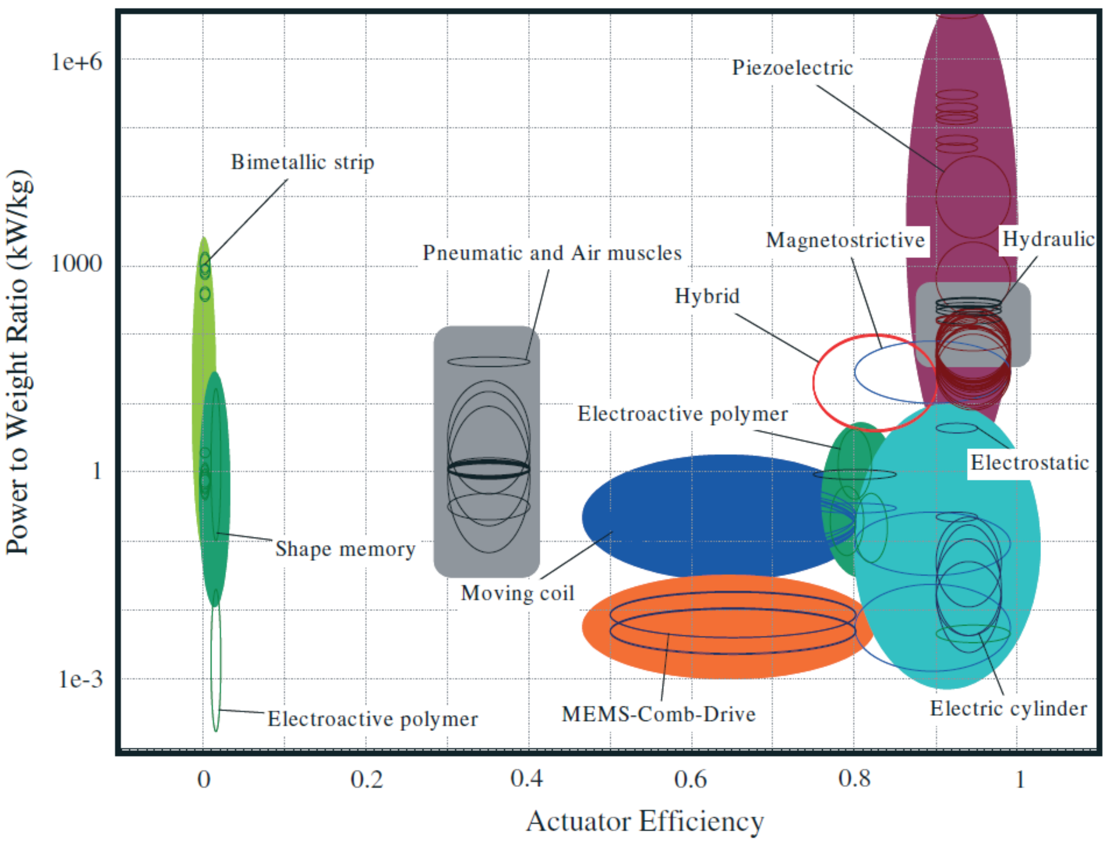
Trade-off needed

What do you need?

Requirements:

Force	5 N
Stroke	2 mm
Frequency	50 Hz
Life	At least 10^8 cycles
Mass	As light as possible





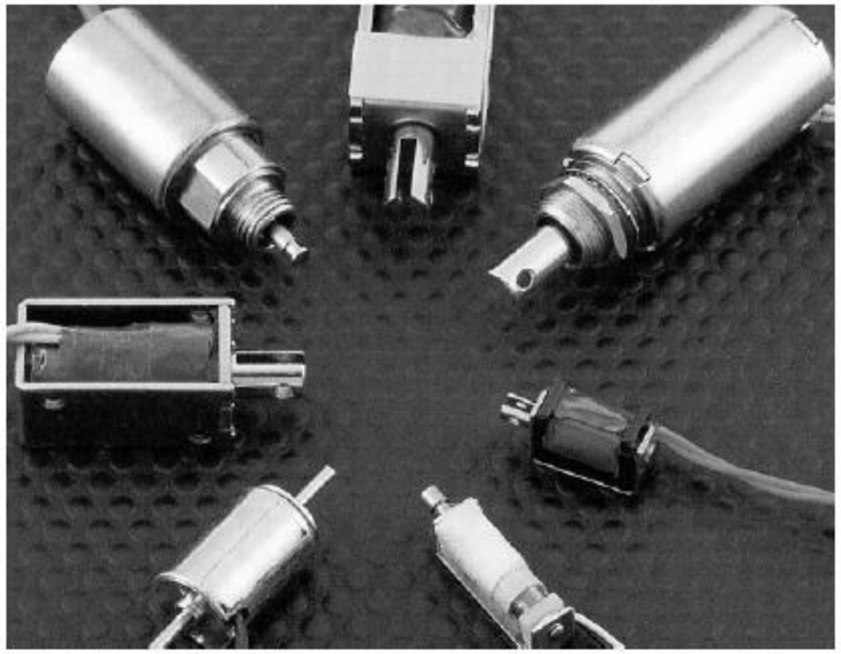
Trade-off needed

What do you need?

Requirements:

Force	5 N
Stroke	2 mm
Frequency	50 Hz
Life	At least 10^8 cycles
Mass	As light as possible

Bicron Solenoid STP 1717-014



Description

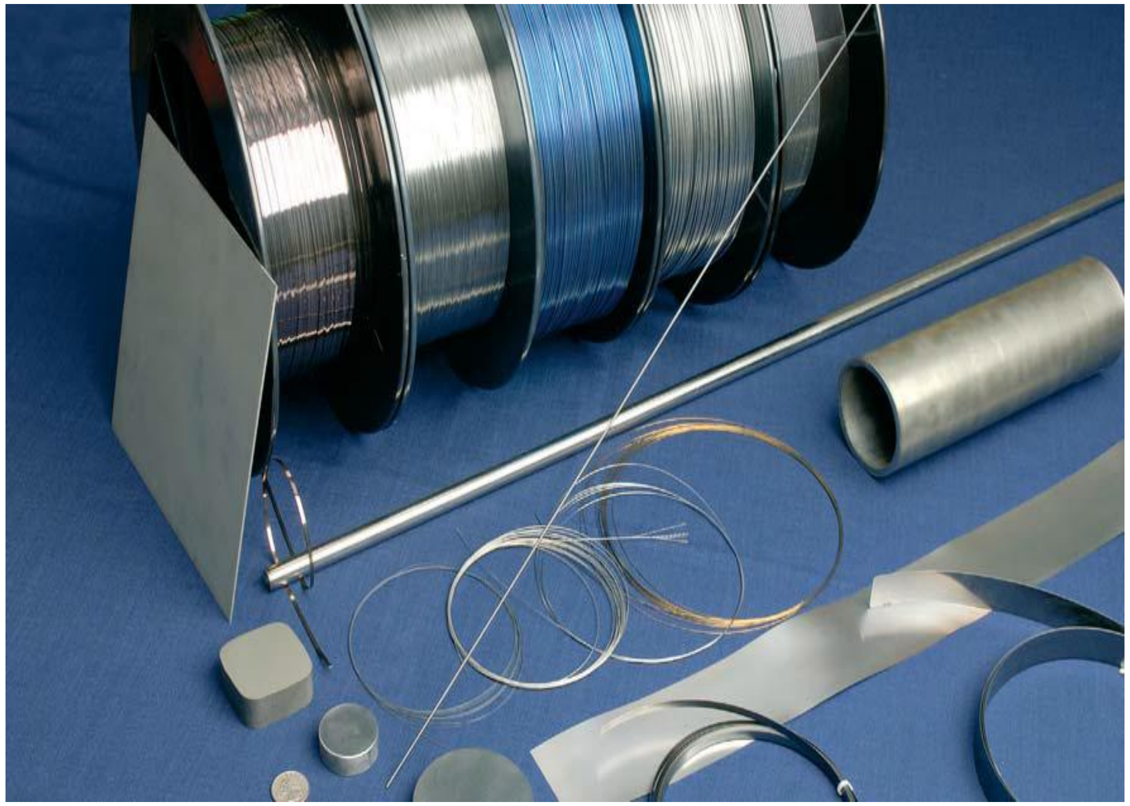
Bicron push-pull solenoids offer high forces in a limited space. Can be can be easily modified to include special features or value added assemblies for non-standard applications.

Characteristics				
Nominal length	0.013	-	0.0135	m
Radius	6.84e-003	-	7.12e-003	m
Maximum stroke	2.98e-003	-	3.1e-003	m
Weight	7.29e-003	-	9.52e-003	kg
Type of power	Electrical			
Voltage required	3			V
Current consumed	0.206			A
Resistance	5.88	-	6.12	Ohms
Performance				
Motion	Linear			
Max linear velocity	0.0149	-	0.248	m/s
Max frequency	*5	-	80	Hz
Max force, compression	13.6	-	14.2	N
Max force, tension	13.6	-	14.2	N
Max power generated	*1.98e-005	-	8.92e-005	kW
Stroke work coefficient	*0.5	-	1	
Cyclic power coefficient	*0.25	-	0.5	
Life	1e+005	-	1e+007	cycles
Derived Properties				
Actuator density	*3800	-	4400	kg/m^3
Actuator modulus	3.314e+004	-	3.59e+004	Pa
Actuator strain	0.22	-	0.24	
Resolution (strain min)	*1e-004	-	0.01	
Actuator stress, compr.	8.54e+004	-	9.63e+004	Pa
Actuator stress, tension	8.54e+004	-	9.63e+004	Pa
Max power density	*1e+004	-	4e+004	W/m^3
Actuator efficiency	*0.5	-	0.8	

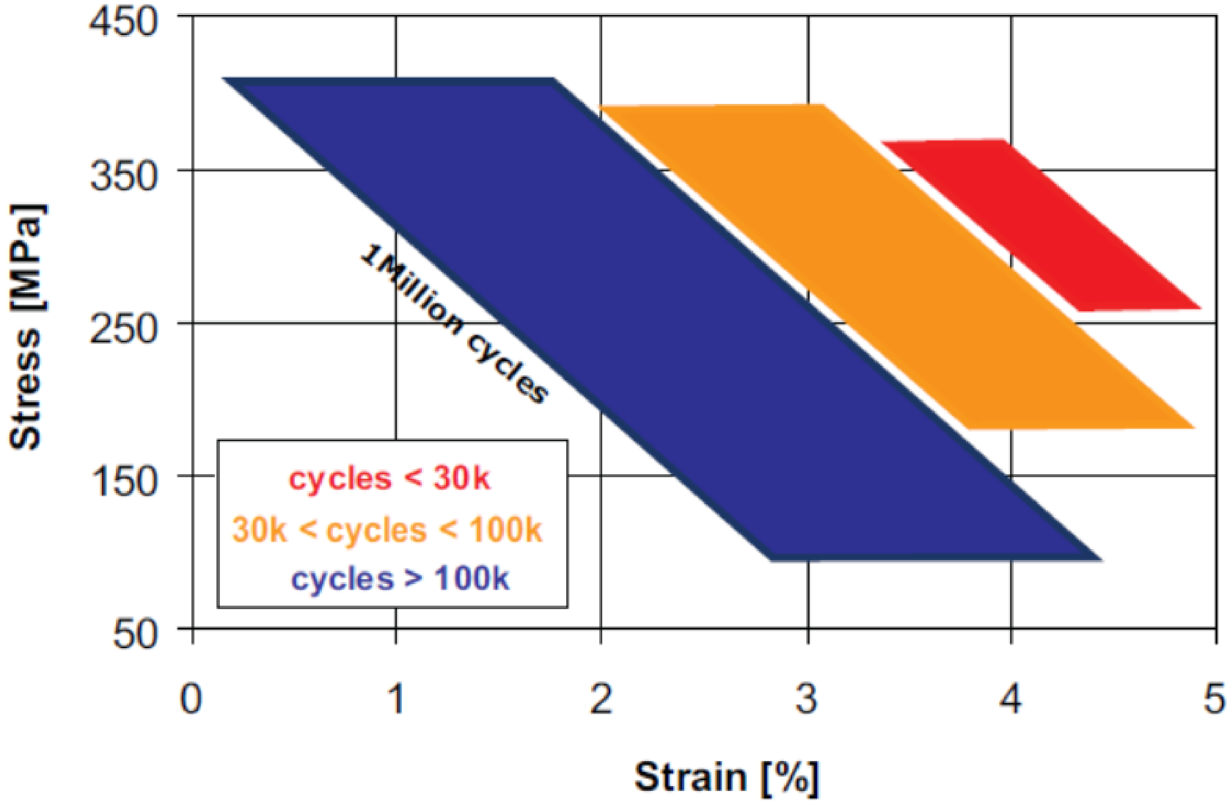
What do you need?

Requirements:

Force	5 N
Stroke	2 mm
Frequency	50 Hz
Life	At least 10^8 cycles
Mass	As light as possible



SMA - SmartFlex®



Innovative actuators

NEW TRANSDUCING PHENOMENON

↓

- **To Master the New Phenomenon.** To identify all the implications of the new transduction until constitutive equations that fully describe the phenomenon are finally arrived at.

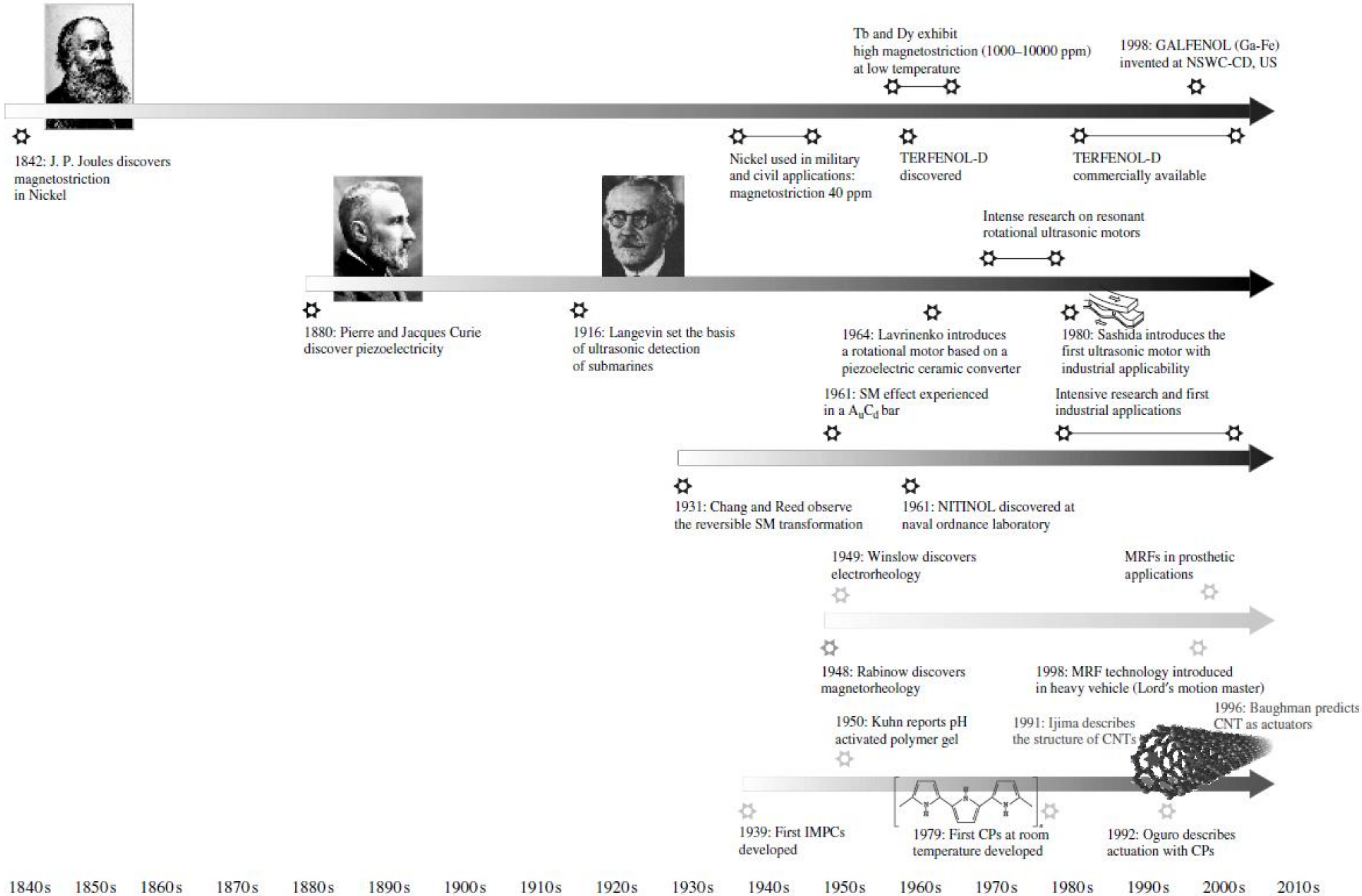
- **To Master the Material Aspects.** The new transduction level (energy out to energy in) may not be appropriate with existing materials. New materials offering enhanced transduction must be engineered.

- **To Develop Appropriate Actuation Concepts.** Even when the transduction phenomenon and the materials are there, actuation concepts may not be apparent. This is true, for instance, of travelling wave ultrasonic motors, which have only recently been developed out of piezoelectric materials.

↓

USABLE ACTUATORS

Actuation principles timeline



Transduction technologies

ELECTRIC ENERGY CONVERTED INTO MECHANICAL DOMAIN

Active: → **Work exchange can be both positive and negative meaning that they can increase or decrease the energy level of the controlled system**

- Piezoelectric Materials
- Shape Memory Materials
- Electro-Active Polymers
- Flexible Fluidic Actuators
- Magneto-strictive Actuators
- Supercoiled Polymer Actuators

Semi-active: → **Work exchange can be only negative or null meaning that they can only dissipate energy during the mechanical interaction with the controlled system**

- Electro Rheological Fluids
- Magneto Rheological Fluids
- Jamming transition phenomenon
- Low melting-point materials

Take home lesson

What kind of actuators do you need?

It depends…

IN EVALUATING THE BEST ACTUATION TECHNOLOGY FOR YOUR APPLICATION YOU SHOULD CONSIDER ALL THE STEPS NEEDED FOR THE ADAPTATION OF THE ENERGY FROM THE SOURCE TO THE FINAL USE

References and further readings

- Pons, J.L., "Emerging Actuator Technologies: A Micromechatronic Approach", John Wiley & Sons, Ltd, England, 2005